Donghwan Rho

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Research Interests

- Implementing Language Models Under Homomorphic Encryption
- Mathematical Reasoning in Large Language Models
- Interpretability of Language Models

Education

Ph.D. **Seoul National University**, Mathematical Sciences — Mar. 2021 – Feb. 2026
- Advisor: *Ernest K. Ryu*

BS **Korea University**, Mathematics Education — Mar. 2015 – Feb. 2021
- Graduated 1st in Class
- (Mandatory) Military Service: May 2017 - Jan. 2019
- Teacher's Certificate for Secondary Regular Teacher of Mathematics

Experience

Staff Engineer, Samsung Electronics — Suwon, South Korea, Mar. 2026 -
- Domain QA Models

Research Internship, CryptoLab — Seoul, South Korea, Dec. 2025 - Feb. 2026
- Homomorphic Encryption Optimization Team

Workshop Coordinator, SNU-KAIST ML Theory Workshop — Gangneung, South Korea, Aug. 2024
- Assisted in planning the schedule, selecting the venue, and managing workshop programs.

Teaching Practicum — Seoul, South Korea, Apr. 2020 - May 2020
- Danggok high school

Militery Service, Auxiliary Police — Seoul, South Korea, May 2017 – Jan. 2019
- Assisted in maintaining public order and security.

Publications and Priprints

*: Equal Contribution

[3] **Bridging LLMs and Symbolic Phonological Rules: RLVR for the Korean Word-Chain Game** — Oct. 2025
Donghwan Rho
Under review
Paper

[2] **Traveling Salesman-Based Token Ordering Improves Stability in Homomorphically Encrypted Language Models** — Oct. 2025
Donghwan Rho*, Sieun Seo*, Hyewon Sung, Chohong Min, Ernest K. Ryu
Under review
Paper

[1] **Encryption-Friendly LLM Architecture** — Oct. 2024

Donghwan Rho*, Taeseong Kim*, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest K. Ryu, Jung Hee Cheon
International Conference on Learning Representations (ICLR) 2025
Paper Code

Talks & Presentations

Symposium of the Research Institute of Mathematics, Seoul National University, Talk — Nov. 2025

- Title: Implementing Efficient Language Models under Homomorphic Encryption
  Site link, Blog post (Thanks to Yossi Frenkel), Slides

Korean Mathematical Society Annual Meeting, Talk — Oct. 2025

- Title: Homomorphically encrypted language models stabilized with TSP-sorted tokens
  Link

SNU-Samsung Electronics Collaboration Symposium, Poster Presentation — Aug. 2024

- Title: Encryption-Friendly LLM Architecture

SNU-KAIST ML Theory Workshop, Poster Presentation — Aug. 2024

- Title: Encryption-Friendly LLM Architecture

Honors and Awards

Outstanding TA, Seoul National University Link — July 2025

Introduction to Computer Programming and Artificial Intelligence for Scientists

University Students Contest of Mathemaitcs, Silver Prize Link — Dec. 2018

Academic Excellence Scholarship, Korea University — Aug. 2015

Teaching

Peer Tutoring in Basic Sciences, Lead Teaching Assistant — Mar. 2024 - Feb. 2026

- Represented section TAs to ensure the smooth operation of the peer tutoring program.
- Oversaw tutorings in Mathematics, Physics, Chemistry, Biology, and Statistics.

Basic Calculus, Teaching Assistant — Sep. 2021 - Feb. 2026

- Matched tutors and tutees for the *Basic Calculus* peer tutoring program.
- Managed program logistics to ensure efficient operation.

SNU Pre-College Program, Head TA, Instructor

- Managed logistics of the mathematics program for freshman undergraduate.
- Gave lectures.
- Winter 2025, Winter 2024, Winter 2023, Winter 2022

Introduction to Computer Programming and Artificial Intelligence for Scientists, Instructor — Spring 2025

- English Course
- Selected as an outstanding TA

Topics in Machine Intelligence — Fall 2024

- Topics: Continuous-depth neural networks, Diffusion Models, Large Language

Models, Vision Language Models, State-space Models
- English Course
- Graduate Course

Mathematical algorithms II Fall 2022
- Topics: Reinforcement Learning, Diffusion Models, Natural Language Processing
- English Course
- Graduate Course

Calculus 2 Practice, Instructor
- Fall 2021, Fall 2025

Calculus 1 Practice, Instructor
- Spring 2024, Fall 2023, Spring 2023, Spring 2022, Spring 2021

Technical Skills

Languages: Korean (Native), English (Fluent)

Programming Languages: Python, C++

Tools & Technologies: LaTeX, GitHub